

UNIVERSITY OF SOUTHERN CALIFORNIA

Statistical genetics on All of Us

Lab 1: a phenotype, a polygenic index, and a regression on v9 data

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What we will do today

Step	What we do	Script (in scripts/)
0	Start a cloud environment, get the code, fill in settings	00_preflight.sh
1	Copy chromosome-22 genotypes and ancestry data to the VM	01_fetch_genotypes.sh
2	Build a height phenotype from the CDR	02_build_phenotype.sh
3	Summarize it: a summary table and two figures	03_explore_phenotype.sh
4	Build a polygenic index (PGI); check its histogram	04_build_pgi.sh
5	Regress height on the PGI and covariates	05_pgi_regression.sh
6	Save the run; download aggregate results	06_save_run.sh

For each step: a slide on what it does, then a  **Run it** slide with the commands and the code. We run each step together.

CDR: Curated Data Repository, the All of Us database, in Google BigQuery. PGI: polygenic index, one number per person built from many genetic variants. PLINK: a standard program for genotype files.

All of Us data use rules

Rule (policy)	In this lab
Participant-level data stays in the Workbench: no copying, downloading, or screenshots. (Data User Code of Conduct)	Person-level files live in <code>lab_data/</code> and <code>work/</code> only.
No count of 1–20 participants may be shared, directly or by calculation. (Data and Statistics Dissemination Policy)	Counts print rounded to 100. Step 6 checks <code>results/</code> .
Downloads are monitored; downloading participant-level data triggers an alert. (Egress Alert Policy)	Downloads come from <code>results/</code> only.
No participant data or counts under 20 in code on GitHub. (Researcher FAQ)	<code>.gitignore</code> excludes data and output folders.
Notify the program 2 weeks before a publication or presentation. (Publication and Presentation Policy)	For your own projects.

Full text: support.researchallofus.org (Policies). Article numbers are listed in the cheat sheet.

Setting up your cloud environment

In the workshop workspace: **Apps** tab → **+ New app instance** → **JupyterLab**.

Setting	Choose	Why
Environment	Standard VM	The Spark cluster option is for Hail.
Machine	4 CPUs, 16 GB memory	Enough for one chromosome.
Disk	100 GB	Today uses under 5 GB. Disks grow, never shrink.
GPU	None	Nothing today uses a GPU.
Autostop	On, 1 hour	Pauses the VM after 1 idle hour.

The creation page shows the hourly cost of the choices. A running VM bills by the hour; a paused VM bills only for its disk.

All of Us support: “Cloud Apps in the Researcher Workbench”; “Getting Started and What to Know About Costs.”

Autostop: when to change it

Autostop pauses the VM when it has been idle. The VM counts as **idle** when, for 15 minutes,

- average CPU use is below 50%, and
- no browser tab or terminal connection is active.

Keep autostop on (1–4 hours) for interactive work, like today.

Set it longer than the job (up to 14 days) when a job runs for hours with low CPU use:

- a single-threaded program on a 4-CPU VM (1 busy CPU = 25% use);
- a large copy from a bucket, which waits on the network, not the CPU;
- a script that waits on a long BigQuery query.

Otherwise the VM can pause in the middle of the job. **Turn autostop off** only if you will stop the VM yourself.

GitHub: what it is and how we use it

Git records versions of the files in a folder. Each saved version is a *commit*.

GitHub hosts a copy of that folder (a *repository*) online, so others can copy it.

We use it for three things:

1. **Getting the code onto the VM:** `git clone` once, `git pull` for updates.
2. **Recording which code produced a result:** step 6 writes the commit ID into `results/run_manifest.txt`.
3. **Running identical code** in every participant's workspace.

GitHub is outside the Workbench. A repository holds code, settings templates, and documentation. It never holds data, results, filled-in settings, or notebook outputs.

.gitignore: keeping data out of GitHub

A .gitignore file lists files and folders that git never tracks. This repository's:

```
config.sh          # your filled-in settings
lab_data/          # copied-in data (person-level)
work/              # intermediate files (person-level)
results/           # aggregate outputs (reviewed and downloaded by hand)
runs/              # saved archives
.ipynb_checkpoints/ # notebook autosaves
```

Practices that carry over to any project:

- Put data and outputs in folders, and ignore the folders by name.
- Commit a settings template (config.example.sh); ignore the filled-in copy (config.sh).
- Clear notebook outputs before committing (JupyterLab: Edit → Clear All Outputs). A saved .ipynb file stores its outputs.
- Before git push, run git status and git diff -cached to see exactly what will be sent.

■ Run it – Step 0: get the code and fill in settings

JupyterLab Launcher → **Terminal**, then:

```
git clone https://github.com/jeffreyswigert/aou-statgen-lab.git
cd aou-statgen-lab
cp config.example.sh config.sh      # your own copy of the settings
nano config.sh                      # fill in two values; Ctrl-O saves, Ctrl-X exits
git check-ignore -v config.sh       # confirms git ignores config.sh
bash scripts/00_preflight.sh
```

The two values to fill in (shown on screen):

```
export WORKSHOP_BUCKET="gs://..." # the workshop workspace's bucket
export BILLING_PROJECT="..."       # the Google project billed for queries and copies
```

Expected: the last line reads Preflight passed. If a value is missing, preflight names it and stops.

Code: scripts/00_preflight.sh; scripts/common.sh (settings loaded by every script)

Where today's data are

1. The CDR, in BigQuery: `wb-silky-artichoke-2408.C2025Q4R6`

Tables `measurement`, `person`, `concept` → the height phenotype (step 2)

2. The All of Us controlled bucket: `gs://vwb-aou-datasets-controlled/v9/`

`wgs/short_read/snpindel/aux/ancestry/ancestry_preds.tsv`

predicted genetic ancestry and 16 principal components per person (step 1)

3. The workshop bucket: `$WORKSHOP_BUCKET`

`genotypes/chr22_hm3.{bed,bim,fam}`: chromosome-22 genotypes, HapMap3 variants only (step 1)

`pgi_workshop/height_weights.txt`: PGI weights from a published height GWAS (step 4)

The controlled bucket is **requester-pays**: each copy names who pays (`-billing-project`). PLINK reads only files on the VM's disk, so step 1 copies the files there.

Where the genotypes come from

- All of Us publishes whole-genome genotypes as one PLINK file set per chromosome: `.../v9/wgs/short_read/snpindel/acaf_threshold/plink_bed/`
- The chromosome-22 .bed file is about 250 GB. All of Us does not publish a smaller HapMap3 subset.
- **HapMap3 variants:** about 1.2 million common variants from the International HapMap Project. Most GWAS report them, so they are a standard set for building PGIs.
- Before class, the instructor kept only the HapMap3 variants on chromosome 22 (~2 GB) and put the result in the workshop bucket.

How (scripts/prep/make_hm3_subset.sh, run once):

```
# keep variants whose position and alleles match the HapMap3 list
plink2 --bfile acaf --extract extract.txt --make-bed --out tmp_hm3
# rename All of Us IDs (chr22:16050075:A:G) to rsIDs, to match the GWAS weights
plink2 --bfile tmp_hm3 --update-name rename.txt --make-bed --out chr22_hm3
```

■ Run it – Step 1: copy the genotypes and ancestry file

```
bash scripts/01_fetch_genotypes.sh      # 1-3 minutes
```

What it runs (scripts/01_fetch_genotypes.sh):

```
gflags=(--billing-project="$BILLING_PROJECT")      # who pays for the copy
for ext in bed bim fam; do
  gcloud "${gflags[@]}" storage cp "$GENO_SRC/chr${CHROM}_hm3.$ext" lab_data/
done
gcloud "${gflags[@]}" storage cp "$ANC_SRC" lab_data/ancestry_preds.tsv
```

Then look at the first lines of each file:

```
head -n 2 lab_data/chr22_hm3.fam
head -n 2 lab_data/chr22_hm3.bim
head -n 1 lab_data/ancestry_preds.tsv | tr '\t' '\n'      # one column name per line
```

Expected: chr22: ... variants staged, about 17,000 variants. Everyone's count is the same.

The files we just copied

File (one line =)	Columns	Note
chr22_hm3.fam (person)	FID, IID, father, mother, sex, phenotype	FID is 0 for everyone. IID is the person ID.
chr22_hm3.bim (variant)	chromosome, ID, cM, position, allele 1, allele 2	ID is an rsID; positions are GRCh38.
chr22_hm3.bed	genotypes, binary	Never open as text.
ancestry_preds.tsv (person)	research_id, ancestry_pred, probabilities, pca_features, ancestry_pred_other	research_id is the CDR person_id. pca_features holds 16 PCs in one bracketed text field.

Made-up lines, showing the layout (one person ID, three column names: IID, person_id, research_id):

```
0 1000010 0 0 2 -9 <- .fam
22 rs1234567 0 17054321 A G <- .bim
1000010 eur [0.01, ...] [-0.0091, 0.0152, ...] eur <- ancestry_preds.tsv
```

The phenotype data: where height comes from

- Height is in the CDR measurement table: one row per measurement, often several rows per person.
- We select rows with `measurement_source_concept_id = 903133`: height measured by the program at enrollment, plus remote self-report.
- We do not select on the standard concept 3036277: it also returns heights from electronic health records.
- Year of birth and sex at birth come from the person table.
- `SAMPLE_MOD=10` keeps people whose `person_id` is divisible by 10, about 1 in 10, so queries run quickly in class. `SAMPLE_MOD=1` keeps everyone.

Concept 903133: All of Us support article 29888188023060 (updated July 2026). Look up other concepts in the public Data Browser, databrowser.researchallofus.org.

The query and the files it writes

The first of two queries (scripts/02_build_phenotype.sh):

```
SELECT person_id, value_as_number
FROM '$CDR_DATASET.measurement'
WHERE measurement_source_concept_id IN (903133)
      AND value_as_number IS NOT NULL
      AND MOD(person_id, 10) = 0
```

The second gets year_of_birth and sex_at_birth from person. Results go to work/ (made-up values):

```
work/raw_pheno.csv
person_id,value_as_number
1000010,172.4
1000010,172.0
1000020,158.9
```

```
work/raw_person.csv
person_id,year_of_birth,sex_at_birth
1000010,1961,Female
1000020,1983,Female
```

Note the two rows for person 1000010 in raw_pheno.csv: the next step reduces them to one.

■ Run it – Step 2: build the phenotype

```
DRY_RUN=1 bash scripts/02_build_phenotype.sh    # prints the two queries; runs nothing
bash scripts/02_build_phenotype.sh              # runs the queries, then cleans (1-2 min)
```

The cleaning (scripts/build_pheno.py, steps 1–5, lightly simplified):

```
v = statistics.median(values)                # 1: each person's median
if not LO <= v <= HI: continue                # 2: drop outside 100-250 cm
sex = sex if sex in ('Male', 'Female') else 'other' # 3: sex at birth
age = 2026 - int(yob)                         # 4: age
z = (v - mu) / sd                             # 5: standardize within sex
```

Output work/aou_pheno.tsv, one row per person: person_id value value_z age sex.

Expected (counts rounded to the nearest 100):

```
measurement records pulled    ~...
people with any measurement    ~...
removed: outside 100-250 cm    ~...
people in analysis file        ~...
```

Cleaning and exploratory data analysis: the summary table

Input: work/aou_pheno.tsv. Made-up rows, showing the columns:

person_id	value	value_z	age	sex
1000010	172.2	1.3172	65	Female
1000020	158.9	-0.4127	43	Female

Check it from the terminal (output stays on your screen):

```
head -n 3 work/aou_pheno.tsv | column -t      # columns line up
wc -l work/aou_pheno.tsv                      # rows + 1 header line
cut -f 5 work/aou_pheno.tsv | sort | uniq -c   # rows per sex
```

Outputs in results/: aou_pheno_summary.txt (the table), aou_pheno_hist.png, aou_pheno_kde_sex.png.

The table follows Stata's summarize layout, with two changes: Obs is rounded to the nearest 100, and percentiles replace Min and Max (each is one person's value).

Code: scripts/pheno_eda.py - sum_row() makes each table row

■ Run it – Step 3: the summary table and figures

```
bash scripts/03_explore_phenotype.sh
cat results/aou_pheno_summary.txt
```

Expected layout (made-up numbers):

Variable	Obs	Mean	Std. dev.	p25	Median	p75
height	~X,X00	169.10	9.80	162.0	168.9	176.2
Male	~X,X00	175.90	7.40	171.0	175.9	180.8
Female	~X,X00	162.40	6.90	157.8	162.3	167.0

Checks:

- US adult means: about 176 cm (men) and 162 cm (women); SDs 7–8 cm.
- In the histogram, a spike or a second hump usually means mixed units or a wrong concept ID.

Open the figures by double-clicking them in the JupyterLab file browser, folder `results/`.

Building the PGI

For person i , the PGI adds up, over variants j , the variant's weight times the person's number of copies of the effect allele:

$$\text{PGI}_i = \sum_j w_j g_{ij}, \quad g_{ij} \in \{0, 1, 2\}$$

The weights w_j are per-allele effect estimates from a published height GWAS (genome-wide association study).

File	Contents
lab_data/chr22_hm3.{bed,bim,fam}	genotypes (in)
lab_data/height_weights.txt	weights (in): rsid effect_allele weight
work/aou_pgi.sscore	one row per person (out); SCORE1_SUM is the PGI
work/aou_pgi.log	PLINK's log (out), including the match count

Where a genotype is missing, PLINK uses the variant's average allele count.

The PLINK command

From scripts/04_build_pgi.sh (with CHROM=22):

```
p2 --bfile lab_data/chr22_hm3 \  
  --score lab_data/height_weights.txt 1 2 3 header cols=+scoresums list-variants \  
  --out work/aou_pgi
```

Part	Meaning
p2	plink2 with our CPU and memory settings (scripts/common.sh)
-bfile lab_data/chr22_hm3	read chr22_hm3.bed, .bim, .fam
-score FILE 1 2 3	weights file; variant ID in column 1, effect allele in 2, weight in 3
header	the first line of the weights file is column names
cols=+scoresums	also write each person's sum (SCORE1_SUM)
list-variants	write the variants used to work/aou_pgi.sscore.vars
-out work/aou_pgi	prefix for the output files

■ Run it – Step 4: build the PGI

```
bash scripts/04_build_pgi.sh          # weights, PLINK, QC histogram
grep -E 'variants processed|skipped' work/aou_pgi.log  # the match count
wc -l < lab_data/height_weights.txt    # weight rows + 1 header line
```

The match count. variants processed is the number of weight rows that found their variant in the genotype file. PLINK skips unmatched rows without an error.

- Processed close to the number of weight rows: expected.
- Processed near zero: the variant IDs do not match (for example, rsIDs in the weights and chr22:16050075:A:G in the .bim), or the column numbers are wrong.

Expected: everyone gets the same processed count; the instructor gives the rehearsal value.

QC on the PGI: a histogram

Step 4 standardizes the PGI and plots its histogram with a normal curve for reference:

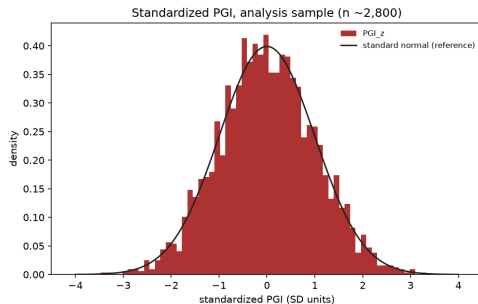
$$\text{PGI_}z_i = \frac{\text{PGI}_i - \text{mean}(\text{PGI})}{\text{SD}(\text{PGI})}$$

```
pgi_z = (raw - raw.mean()) / raw.std()  
ax.hist(pgi_z, bins=60, density=True)
```

Expected: one hump centered at 0, close to the curve; skewness near 0 and kurtosis near 3 (in results/aou_pgi_summary.txt).

A second hump, a spike, or a long tail points to the inputs: many missing genotypes, or a few very large weights.

Code: scripts/pgi_eda.py



Example output of this code on simulated data (not All of Us data).

The regression

$$\text{height_}z_i = \alpha + \beta \text{PGI_}z_i + \gamma \text{age}_i + \delta \text{female}_i + \sum_{k=1}^5 \theta_k \text{PC}_{ik} + \varepsilon_i$$

Variable	Definition	File (step)
height_z	(height – mean for that sex) / SD for that sex	work/aou_pheno.tsv (2)
PGI_z	(PGI – mean) / SD, over the people in the model	work/aou_pgi.sscore (4)
age	2026 – year of birth	work/aou_pheno.tsv (2)
female	1 if Female, 0 if Male	work/aou_pheno.tsv (2)
PC ₁ –PC ₅	first 5 genetic principal components	lab_data/ancestry_preds.tsv (1)

Estimated by ordinary least squares (OLS) on the 1-in-10 sample.

What the regression reports

- β : height and the PGI are both in SD units, so β is the difference in height, in SDs, per 1 SD of PGI, with age, sex, and the PCs held fixed.
- **The PCs** summarize broad genetic ancestry. Including them keeps ancestry-related differences in height from being attributed to the PGI.
- **Incremental** $R^2 = R^2$ of the full model $- R^2$ of the same model without PGI_z. It is the share of height variance the PGI explains beyond the covariates.
- **Scope of today's estimate**: one chromosome, a 1-in-10 sample, and OLS, which treats participants as unrelated. A full PGI uses all 22 chromosomes.

■ Run it – Step 5: the regression

```
bash scripts/05_pgi_regression.sh
cat results/aou_pgi_regression.txt
```

The fit (scripts/pgi_regression.py):

```
X[:, 0] = (X[:, 0] - X[:, 0].mean()) / X[:, 0].std() # PGI -> PGI_z
beta, se, r2_full = fit(X) # height_z on PGI_z, age, female, PC1-PC5
_, _, r2_cov = fit(X[:, 1:]) # the same model without PGI_z
```

Expected layout (made-up numbers):

term	estimate	SE	t
PGI_z	0.0XXX	0.00XX	X.XX
...			
R2 with PGI	= 0.0XXX		
R2 covariates	= 0.0XXX		
incremental R2	= 0.00XX		

After class: delete the five PC names from COVARs in scripts/pgi_regression.py, re-run step 5, and compare the PGI_z row.

Saving results and downloading them

Folder	Contents	Can it leave the Workbench?
lab_data/	copied-in genotypes, ancestry file, weights	No
work/	person-level files: phenotype, PGI scores	No
results/	summary tables (.txt), figures (.png)	Yes, after the check and your review
runs/	archive of the run (copied to the bucket)	No

To download a file: open it and read it; then JupyterLab file browser → right-click → Download.

- Figures and free text are not checked by the script; review them yourself.
- Copying out with `gcloud storage cp` to storage outside the Workbench is blocked.
- A large summary file needs an approved exception first (support article 48941559021716).

What step 6 does

`scripts/06_save_run.sh`:

1. **Checks** `results/` (`scripts/check_disclosure.py`) for counts of 1–20, direct or from the difference of two counts, and for person-level files. If it finds any, it stops and names the file.
2. **Writes** `results/run_manifest.txt`: the git commit ID, the settings, the PLINK version, and checksums of the input files.
3. **Packs** `config.sh`, `scripts/`, `work/`, and `results/` into one archive in `runs/`.
4. **Copies** the archive to `$WORKSHOP_BUCKET/statgen-lab/` and lists it to confirm.

The archive holds person-level files, so it stays in the workspace bucket. It is never downloaded.

■ Run it – Step 6: save, download, stop

```
bash scripts/06_save_run.sh
cat results/disclosure_report.txt           # what the check found
grep code_version results/run_manifest.txt  # the commit that produced this run
gcloud storage ls "$WORKSHOP_BUCKET/statgen-lab/" # the archive is in the bucket
```

Expected: Findings ... none found, then the archive's bucket path.

If the check stops the save: it prints the file and the count. Round, combine, or remove that cell, then re-run step 6.

Download one file: file browser → results/ → right-click aou_pgi_hist.png → Download.

Stop the VM: Apps tab → your JupyterLab app → **Pause**.

Code: scripts/06_save_run.sh; scripts/check_disclosure.py

All the commands, in order

```
git clone https://github.com/jeffreyswigert/aou-statgen-lab.git && cd aou-statgen-lab
cp config.example.sh config.sh && nano config.sh # WORKSHOP_BUCKET, BILLING_PROJECT
bash scripts/00_preflight.sh # check tools and settings
bash scripts/01_fetch_genotypes.sh # genotypes + ancestry file -> lab_data/
bash scripts/02_build_phenotype.sh # CDR queries, one row per person -> work/
bash scripts/03_explore_phenotype.sh # summary table, 2 figures -> results/
bash scripts/04_build_pgi.sh # PGI, match count, histogram -> work/, results/
bash scripts/05_pgi_regression.sh # the regression -> results/
bash scripts/06_save_run.sh # check, manifest, archive to the workspace bucket
```

Settings to change (in config.sh, or before one command):

- PHENO=ld1 – another phenotype (bash scripts/02_build_phenotype.sh list)
- SAMPLE_MOD=1 – everyone instead of 1 in 10
- CHROM=21 – another chromosome (after making its subset with scripts/prep/make_hm3_subset.sh)

Cheat sheet: cheatsheet.pdf in the repository.